PEARLS

AQI Predictor

Project architecture, data science, operations, and implementation review

Prepared from the repository state
8 September 2026

End-to-end air-quality forecasting for Hyderabad, Pakistan
24h • 48h • 72h multi-horizon PM2.5 forecasts

Executive Summary

What the project delivers

Pearls AQI Predictor is an end-to-end air-quality forecasting platform for Hyderabad, Pakistan. It collects hourly weather and pollutant observations from Open-Meteo, stores data in MongoDB Atlas, builds time-series features, trains multi-output regressors, registers the selected model in remote MLflow through DagsHub, and exposes the live forecast through a Streamlit dashboard.

CORE OUTCOMES

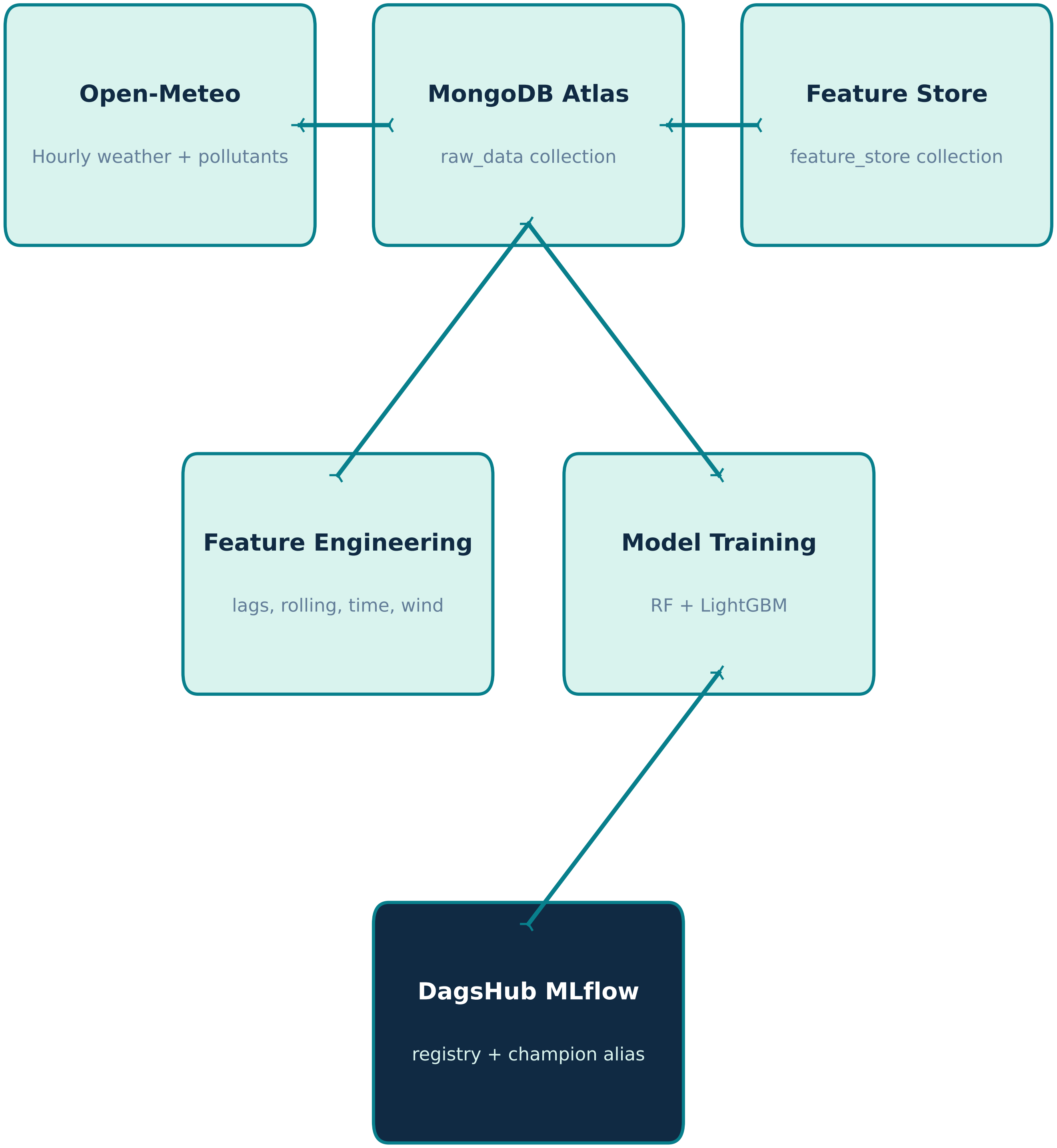
- Forecast horizons: PM2.5 at 24, 48, and 72 hours ahead.
- Automated hourly ingestion and feature refresh, plus daily training and promotion.
- Candidate comparison using MAE, RMSE, and R2 with a temporal validation split.
- A champion model alias makes the dashboard's production model explicit and replaceable.
- Dashboard views include current conditions, trends, forecast outlook, health guidance, and downloads.

CURRENT SCOPE

The repository is configured for a single default location: 25.3548, 68.3585. It is an operational prototype with external-service dependencies, not a self-contained offline demonstration. Secrets and live data are intentionally not included in the report.

System Architecture

Data, feature, model, and presentation layers



The Streamlit app reads the champion model and feature data to render the live user experience. GitHub Actions invokes the hourly and daily pipelines on a schedule.

Data Engineering

How raw observations become model-ready records

INGESTION

- Open-Meteo archive APIs provide temperature, humidity, wind speed/direction, PM2.5, PM10, CO, NO2, SO2, O3, and dust.
- The first run requests up to 730 days; later runs request a short recent window and de-duplicate timestamps before insertion.
- MongoDB URI validation occurs before the client is created; records are restricted to timestamps that have already passed.

FEATURE ENGINEERING

- Log-transformed PM2.5, lags from 1 to 48 hours, and rolling means/standard deviations from 3 to 24 hours.
- Pollutant memory for PM10 and NO2; wind converted to x/y vectors; cyclical hour and month encodings.
- Stagnation index, weekend flag, and 24-hour temperature/humidity changes capture context beyond raw pollutant values.
- Targets are future PM2.5 values at 24, 48, and 72 hours. Incremental updates use a 30-day source window and a 14-day write window.

Modeling and MLOps

Training, evaluation, and release control

TRAINING FLOW

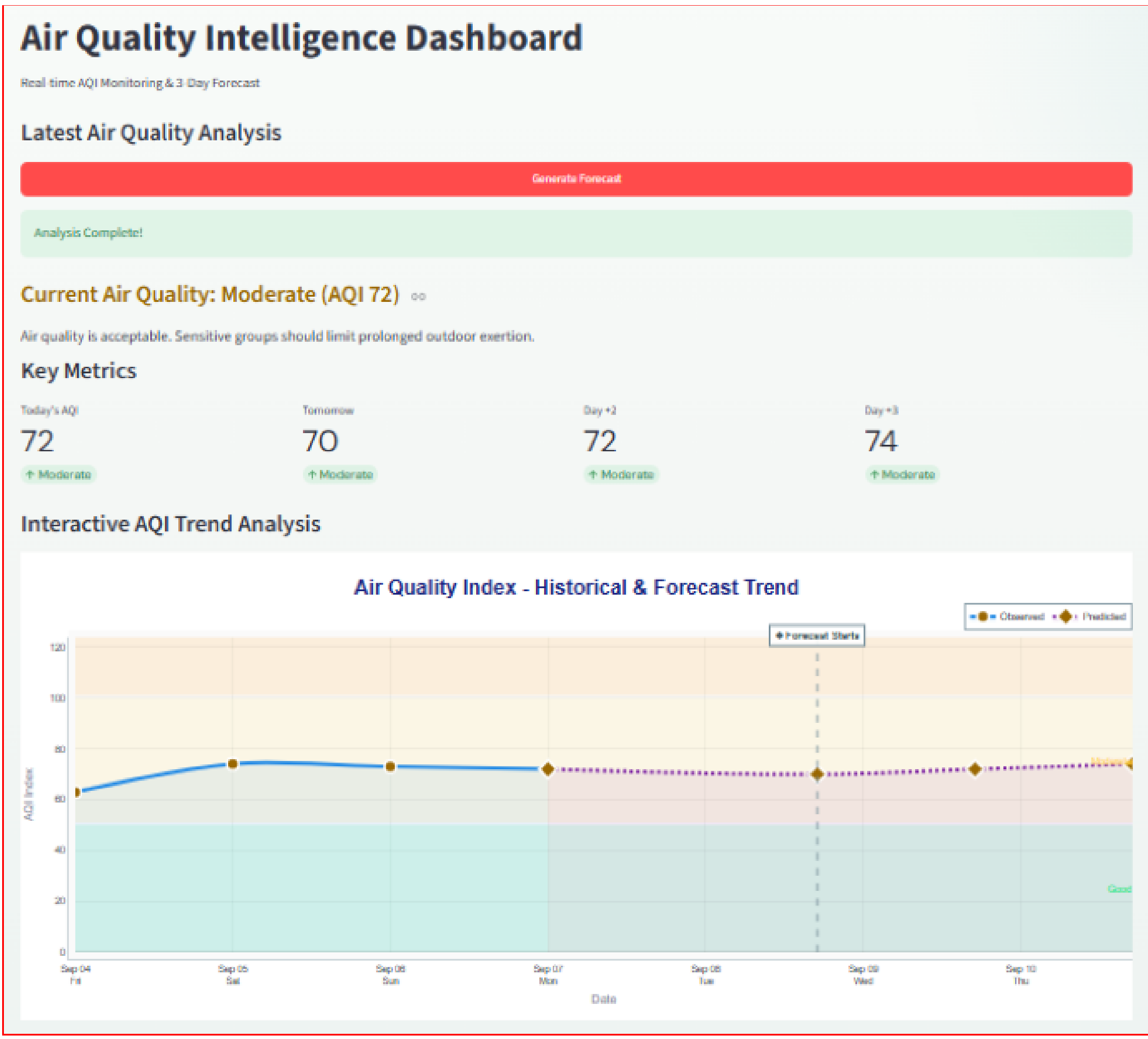
- Feature records are sorted chronologically, coerced to numeric values, and split 80/20 without shuffling.
- Random Forest and LightGBM are trained as MultiOutputRegressor models for the three forecast horizons.
- Predictions are clipped at zero because negative PM2.5 values are invalid.
- The winner is selected by highest average R2, with average RMSE as the tie-breaker, then retrained on all available data.

GOVERNANCE

- MLflow logs the winning model type, feature count, average MAE/RMSE/R2, and the serialized registered model.
- The model registry is named AQI_MultiOutput_Predictor; the dashboard loads the champion alias.
- Promotion compares the newest model against the current champion by average R2, while also refreshing models at least three days old.

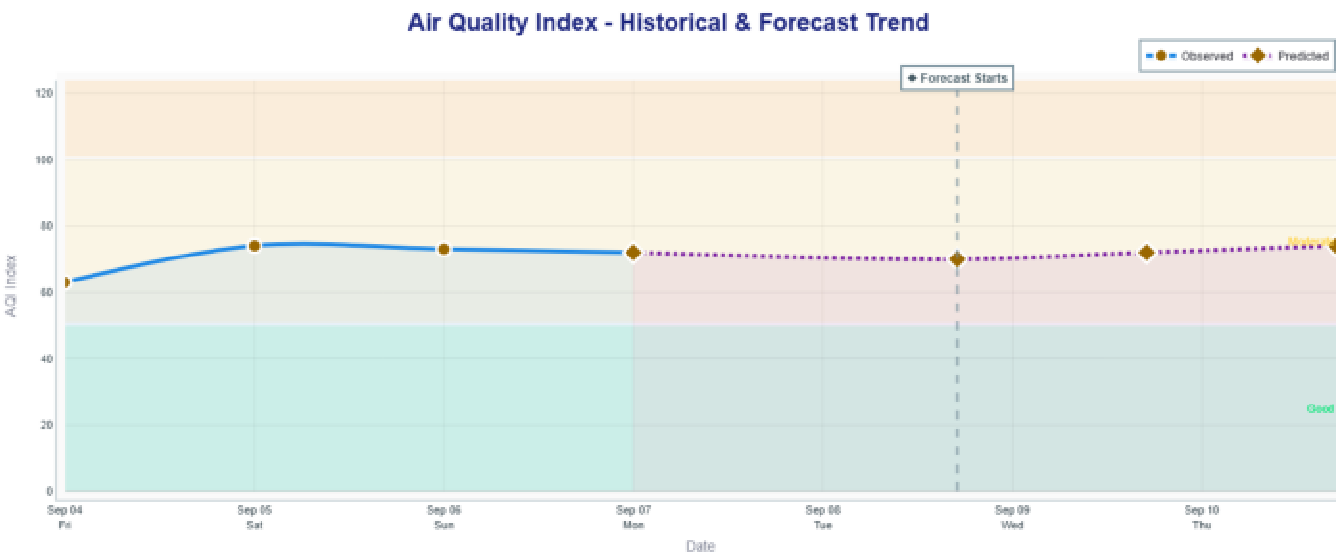
Dashboard Experience

Streamlit dashboard preview included in the repository



Trend and Insight Views

Forecast trend visualization included in the repository



Observed Forecast Snapshot

Dashboard values captured on 7 September 2026

RECENT CONDITIONS

- Past 7-day average PM2.5: 18.18 μg/m³; average PM10: 41.26 μg/m³.
- Average temperature: 28.5 °C; average relative humidity: 73.9%.
- Past 4-day AQI: average 70.2, minimum 63, maximum 74; dashboard trend marked as worsening.

FORECAST OUTLOOK

- Next 3-day AQI: average 70.7, minimum 68, maximum 73; overall outlook marked Moderate.
- Predicted PM2.5: 20.15 μg/m³ at 24 hours, 21.78 μg/m³ at 48 hours, and 22.67 μg/m³ at 72 hours.
- Corresponding forecast AQI values: 68, 71, and 73. Last update shown: 2026-09-07 18:00:00.

INTERPRETATION

This snapshot strengthens the report by demonstrating a plausible user-facing decision context: recent AQI is moderate and the forecast rises gradually across the three horizons. It is descriptive evidence of dashboard operation, not a substitute for validation metrics such as MAE, RMSE, R2, confidence intervals, or comparison against a baseline model.

Operations and Delivery

How the project runs in development and automation

LOCAL COMMANDS

- Install dependencies from requirements.txt in a Python 3.12 virtual environment.
- Run python scripts/pipeline_runner.py hourly for extraction and feature engineering.
- Run python scripts/pipeline_runner.py daily for training and model promotion.
- Run streamlit run app/app.py to open the dashboard at localhost:8501.

SCHEDULED AUTOMATION

- Hourly GitHub Actions workflow runs at five minutes past each hour and updates the data pipeline.
- Daily workflow runs at midnight UTC and executes training plus promotion.
- Both workflows use Python 3.12, install CI dependencies, validate required secrets, and support manual dispatch.

Security, Testing, and Gaps

Repository review findings

SECURITY CONTROLS

- MongoDB and DagsHub credentials are loaded from environment variables and GitHub Actions secrets.
- The README explicitly warns against committing .env files, tokens, passwords, or API keys.
- Runtime checks reject missing or malformed MongoDB and filesystem-based MLflow URIs.

VERIFIED TEST SURFACE

- tests/test_pipeline_runner.py covers the mapping from hourly/daily pipeline names to their scripts.
- Additional diagnostic and integration scripts exist for MongoDB and MLflow, but require external services and credentials.

CURRENT GAPS TO ADDRESS

- The daily GitHub Actions workflow contains an indentation defect in its inline Python validation block and should be corrected before relying on scheduled training.
- The XGBoost training function exists, but the active training path sets xgb_model to None and compares only LightGBM and Random Forest.
- There are no focused tests for feature engineering, model evaluation, promotion decisions, or dashboard data loading.
- The project has no selected license, as noted by the README.

Conclusion and Next Steps

A practical path from prototype to stronger production readiness

The repository demonstrates a coherent MLOps workflow: external data is collected and retained, time-aware features are generated, competing models are evaluated, the selected model is registered, and the dashboard consumes a named champion. The strongest next improvements are operational reliability and test depth rather than a new framework.

RECOMMENDED NEXT STEPS

- Fix and run both GitHub Actions workflows with a minimal validation job before the next scheduled execution.
- Add unit tests for feature columns, target alignment, non-negative predictions, and champion promotion rules.
- Persist model metrics by horizon and expose a compact model-health section in the dashboard.
- Add data-quality checks for missing hours, stale records, and unexpected pollutant ranges.
- Define a project license and document the expected MongoDB indexes and retention policy.

Report basis: README.md, app/app.py, scripts/, automation_scripts/, .github/workflows/, tests/, and assets/.